# ML Scaling Cheat Sheet

**Goal:** Choose the right feature-scaling method based on the numeric data and the ML algorithm.

## 1. First: Do I Need Scaling?

**Usually scale:** KNN, K-Means, SVM, Logistic Regression, Linear Regression with regularization, Neural Networks, PCA.
**Usually do not need scaling:** Decision Trees, Random Forests, Gradient-Boosted Trees and other tree-based models (though preprocessing may still be useful for other reasons).

## 2. Which Scaler Should I Use?

| Data situation | Recommended | Why / notes |
|---|---|---|
| Different units or very different ranges | StandardScaler | Good general default when data is reasonably distributed. |
| Known bounded range, e.g. 0–100 | MinMaxScaler | Maps values to a chosen range, commonly 0–1. |
| Strong outliers | RobustScaler | Uses median and IQR, so extreme values have less influence. |
| Features are approximately Gaussian | StandardScaler | Centers at mean 0 and scales to standard deviation 1. |
| Very skewed positive data | Transform first, then scale | Try log1p, Yeo-Johnson or another suitable transformation before scaling. |
| Need values in [-1, 1] | MinMaxScaler | Set feature_range=(-1, 1). |
| Sparse matrix | MaxAbsScaler | Preserves sparsity; useful for sparse numeric features. |
| Tree-based model | No scaling usually | Splits depend on order/thresholds rather than distance or magnitude. |

## 3. The Main Scaling Methods

### *Standardization — StandardScaler*

**Use:** Use when features have different scales and there are no severe outliers.

```
Formula: z = (x – mean) / standard deviation
```

**Example:** Example: age = 20–60 and income = ■20,000–■2,00,000.

### *Min-Max Scaling — MinMaxScaler*

**Use:** Use when you want a fixed bounded range, commonly 0–1.

```
Formula: x' = (x – min) / (max – min)
```

**Example:** Example: percentage, normalized sensor values, image-style bounded features.

### *Robust Scaling — RobustScaler*

**Use:** Use when outliers are significant.

```
Formula: x' = (x – median) / IQR
```

**Example:** Example: income with a few extremely wealthy observations.

### *Max-Abs Scaling — MaxAbsScaler*

**Use:** Useful for sparse data because it does not center the data.

```
Formula: x' = x / max(|x|)
```

**Example:** Example: sparse feature matrices.

## 4. Read the Numeric Data Before Choosing

| Question | What to inspect | Decision hint |
|---|---|---|
| Are ranges very different? | min, max, standard deviation | Consider StandardScaler or MinMaxScaler. |
| Are there extreme values? | boxplot, quantiles, IQR | Try RobustScaler; investigate the outliers first. |
| Is the distribution heavily skewed? | histogram, skewness, quantiles | Consider log1p/Yeo-Johnson before scaling. |
| Is the feature naturally bounded? | domain knowledge, min/max | MinMaxScaler can be convenient. |
| Is the data sparse? | many zeros / sparse matrix | Consider MaxAbsScaler. |
| Are you using trees? | model type | Scaling is usually unnecessary. |

## 5. Important Examples

| Feature | Typical values | Recommended thinking |
|---|---|---|
| Age | 18–80 | StandardScaler is usually fine. |
| Salary | ■15,000–■50,00,000 | Large range; StandardScaler is a good default. If highly skewed, transform first. |
| House price | ■10 lakh–■10 crore | Often skewed; inspect distribution and outliers. |
| Exam score | 0–100 | MinMaxScaler is reasonable if a bounded 0–1 representation is useful. |
| Number of children | 0–10 | Scaling depends on the model; don't scale just because the feature is numeric. |
| Transaction amount | Mostly small, a few huge values | RobustScaler or transformation such as log1p may help. |

## 6. The Most Important Rule: Prevent Data Leakage

Split your data first. **Fit the scaler only on the training set**, then use that fitted scaler to transform both training and test data. Never calculate scaling statistics using the test set.

```
Correct workflow: Train/Test Split → fit scaler on X_train → transform X_train → transform X_test → train model.
```

## 7. Quick Decision Tree

| Start | → | Are you using a tree-based model? |
|---|---|---|
| Yes | → | Usually NO scaling. |
| No | → | Are there strong outliers? |
| Yes | → | Try RobustScaler; inspect/transform outliers. |
| No | → | Is the feature heavily skewed? |
| Yes | → | Consider log1p/Yeo-Johnson, then scale if needed. |
| No | → | StandardScaler is a strong default. |

**Remember:** There is no universal scaler that is always correct. Choose based on the data distribution, outliers, sparsity, and the algorithm. In practice, compare preprocessing choices using cross-validation.